Algebra: Chapter 0 Solutions

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July 2, 2026

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Preface

This document does not seek to represent the perfect solutions manual to the text at hand, and is more of a way of recording my progress through the text. Should you happen to find this, and wish to contribute or point out any logical flaws in the arguments presented, I would be more than happy to adjust my solutions to perhaps work towards establishing this as a suggested solutions. Should this end up being the case, I will update this section accordingly

1 Chapter 1: Preliminaries: Set Theory and Categories

1.1 Naive Set Theory

Exercise 1.1. Locate a discussion of Russell's paradox, and understand it.

Solution There are many discussions of the paradox. The one I found most intuitive or easy to understand was the labelling of sets as such: We say a set is 'abnormal' if the set contains itself ($S \in S$), or we say that a set is 'normal' if it does not contain itself ($S \notin S$). We now define the set X as the set of all normal sets. We ask the question 'is X a normal or abnormal set?'

Suppose it is normal, then it would be contained within X by definition, making X abnormal. If we instead suppose X is abnormal, then X would not be contained in itself, making X normal.

We arrive at a paradox; that is, X is neither normal or abnormal.

□

Exercise 1.2. Prove that if $\sim$ is an equivalence relation on a set S, then the corresponding family $\mathscr{P}_\sim$ defined in §1.5 in indeed a partition of S: that is, its elements are nonempty, disjoint and their union is S.

Solution We show the desired result by considering the definition of the equivalence class of a given in the text

$$[a]_\sim := \{b \in S | b \sim a\}$$

Firstly, its clear that every element $s \in S$ is contained in its own equivalence class. This is due to the reflexivity requirement of the relation ($s \sim s$). This also shows that every equivalence class is non-empty. We now show that for any two equivalence classes $[a]_\sim$ and $[b]_\sim$ are either disjoint or equal.

Suppose the two sets are not disjoint. Then there exists an element $z \in [a]_\sim \cap [b]_\sim$. Let $n \in [a]_\sim$ be arbitrary. Since we have $n \sim z$ and $z \sim b$, we must have that $n \in [b]_\sim$ by transitivity and so $[a]_\sim \subseteq [b]_\sim$. Similarly, for an arbitrary element $m \in [b]_\sim$ we have $m \sim z$ and $z \sim a$ and so $m \in [a]_\sim$ implying that $[b]_\sim \subseteq [a]_\sim$. Combining these two results means that $[a]_\sim = [b]_\sim$ and the two equivalence classes are equal.

Totalling the results thus far, we see that all elements of S are in a equivalence class, and that these equivalence classes are either disjoint or equal, and so $\mathscr{P}_\sim$ indeed forms a partition of S.

□

Exercise 1.3. Given a partition $\mathscr{P}$ on a set S, show how to define a relation $\sim$ on S such that $\mathscr{P}$ is the corresponding partition.

Solution Let P_i denote the disjoint non empty subsets in $\mathscr{P}$. We define an equivalence relation on elements of S by the following: $a \sim b$ if and only if a and b are contained within the same subset P_i. We need only show this defines a valid equivalence relation.

We clearly have that $a \sim a$ vacuously by the definition of the equivalence relation, so the reflexivity requirement is satisfied. Supposing that $a \sim b$, this means that $a, b \in P_i$ and so equivalently we have that $b \sim a$, satisfying symmetry. Suppose on top of this we have that $b \sim c$. Then this must mean that c is also the in same subset P_i, thereby meaning that $a \sim c$, satisfying transitivity, making this a well defined equivalence relation. By construction this relation generates the exact same partition of S that $\mathscr{P}$ does.

□

Exercise 1.4. How many different equivalence relations may be defined on the set $\{1, 2, 3\}$.

Solution Combining the results of the two previous exercises we know that an equivalence relation defines a partition, $\mathscr{P}_\sim$ of a set S by equivalence classes, and that given any partition $\mathscr{P}$ of S, we can define a corresponding equivalence relation that defines this partition. What this means is that this question is as simple as counting all partitions of the set. We have the followings partitions of $\{1, 2, 3\}$:

$$\{\{1,2,3\}\}, \{\{1,2\},\{3\}\}, \{\{1\},\{2,3\}\}, \{\{1,3\},\{2\}\}, \{\{1\},\{2\},\{3\}\}$$

Giving a total of 5 equivalence relations.

$\square$

Exercise 1.5. Give an example of a relation that is reflexive and symmetric but not transitive. What happens if you attempt to use this relation to define a partition on the set?(Hint: Thinking about the second question will help you answer the first one.)

Solution Rather than coming up with a strict rule for the relation, we will simply construct a family of subsets that is not a partition and show how this defines the desired relation. Take $\{1, 2, 3\}$ and consider the following family of subsets:

$$\{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}\}$$

If we define this as equivalence classes, we get that the relation is reflexive due to the first 3 subset in the family. We also have that $1 \sim 2 \implies 2 \sim 1$ since there is a subset containing both elements, and the same is true for $2 \sim 3$ satisfying the symmetry requirement. However we do not have $1 \sim 3$ despite having $1 \sim 2$ and $2 \sim 3$, and so transitivity is not fulfilled. The main observation to take away here is that if whatever relation we are working with fails to satisfy all 3 requirements, it will result in a family of subsets that are not a partition of the original set, and vice versa.

$\square$

Exercise 1.6. Define a relation $\sim$ on the set $\mathbb{R}$ of real numbers by setting $a \sim b \iff b - a \in \mathbb{Z}$. Prove that this is an equivalence relation, and find a 'compelling' description for $\mathbb{R}/\sim$. Do the same for the relation $\approx$ on the plane $\mathbb{R} \times \mathbb{R}$ defined by declaring $(a_1, a_2) \approx (b_1, b_2) \iff b_1 - a_1 \in Z$ and $b_2 - a_2 \in Z$.

Solution For any $a \in \mathbb{R}$ we have that $a - a = 0 \in \mathbb{Z}$ so $a \sim a$, satisfying reflexivity. If $a \sim b$ then $a - b \in \mathbb{Z}$ which implies that $-(a - b) = b - a \in Z$ and so $b \sim a$, satisfying symmetry. If $a \sim b$ and $b \sim c$ then we have that $a - b$ and $b - c$ are both integers. We must then have that $(a - b) + (b - c)$ is also an integer, and this is exactly equal to $a - c$, and so we then have that $a \sim c$, satisfying transitivity and thereby proving this is a valid equivalence relation. The second equivalence relation given in the exercise is exactly the same, just working element wise on each ordered pair, with the workings being analogous to those done already.

In terms of a description, we can think of the first equivalence relation as the real number line being quotiented down to just the interval $[0, 1]$. Similarly, for the second relation this is a quotient of the cartesian plane where it is quotiented down to the unit square $[0, 1]^2$. If one is familiar with topology, they may know that this set endowed with the quotient topology coming from the standard topology on $\mathbb{R}^2$ gives the 2-torus $\mathbb{T}^2$.

1.2 Functions Between Sets

Exercise 2.1. How many different bijections are there between a set S with n elements and itself?

Solution This can be solved by thinking about the different bijections element-wise in the set. We have n choices for where to send the first element in the set, and once this is chosen we will then have $(n - 1)$ choices of where to send our second element. We can continue this process to find the the total number of bijections is equal to

$$n \times (n - 1) \times (n - 2) \times \dots 2 \times 1 = n!$$

We know by construction these are indeed injective and surjective, and we will cover every single possible combination of element-wise pairings this way, confirming this is indeed the total number of bijections.

$\square$

Exercise 2.2. Prove statement (2) in Proposition 2.1. You may assume that given a family of disjoint nonempty subsets of a set, there is a way to choose one element in each member of the family.

Solution The claim we must prove is the following: *f has a right-inverse if and only if it is surjective.*
($\implies$) Suppose $f : A \to B$ has a right-inverse $g : B \to A$ so that

$$f \circ g = \mathrm{id}_B$$

Let $b \in B$ be any arbitrary elements. We then have that $(f \circ g)(b) = \mathrm{id}_B(b) = b$. This then means that $f(g(b)) = b$ and so we have found a suitable element $g(b) \in A$ that f maps to b, implying that f is surjective.

($\Longleftarrow$) Conversely suppose that f is surjective. For every $b \in B$ we consider the fiber of f over b, $f^{-1}(b)$. Since f is not necessarily injective, each fiber can have any number of elements. However, by the definition of a function, we know that each fiber is disjoint, and by the axiom of choice we can choose an $a \in f^{-1}(b)$ from each fiber. We then define a function $g : B \to A$ by setting $g(b) = a$. We then have that

$$(f \circ g)(b) = f(g(b)) = f(a) = b$$

Implying that $f \circ g = \mathrm{id}_B$ and so f has a right inverse.

□

Exercise 2.3. Prove that the inverse of a bijection is a bijection and that the composition of two bijections is a bijection.

Solution Let $f : A \to B$ be a bijection, and denote its inverse $f^{-1} : B \to A$

Injective: Suppose that $f^{-1}(b') = f^{-1}(b'')$. By injectivity of f we must then have that $f(f^{-1}(b')) = f(f^{-1}(b''))$. By the definition of the inverse, which states that the composition of f and its inverse it equal to the identity function, this then implies that $b' = b''$. Therefore, f^{-1} is injective.

Surjective Let $a \in A$. we then have that $f(a) \in B$ and $f^{-1}(f(a)) = a$ so f^{-1} is surjective. We have thus proved that the inverse is also a bijection.

Let $g : C \to A$ be another bijection. We will show that $f \circ g$ is a bijection.

Injective: Let $c', c'' \in C$ and suppose that $(f \circ g)(c') = (f \circ g)(c'')$. We then have that

$$f(g(c')) = f(g(c''))$$

By the injectivity of f we must have that $g(c') = g(c'')$, and then by injectivity of g we must then have that $c' = c''$ showing that $f \circ g$ is injective.

Surjective: Let $b \in B$ be an arbitrary element. The surjectivity of f means $\exists a \in A$ so that $f(a) = b$. Then by surjectivity of g $\exists c \in C$ so that $g(c) = a$. We then observe the following

$$(f \circ g)(c) = f(g(c)) = f(a) = b$$

Showing that $f \circ g$ is also surjective and thus is a bijection.

□

Exercise 2.4. Prove that 'isomorphism' is an equivalence relation (on any set of sets).

Solution We show that each requirement of an equivalence relation holds, which comes almost immediately from the previous exercises.

Reflexive: For any set A we must have $A \cong A$ as the identity function on A, $\mathrm{id}_A : A \to A$ is a bijection from A to A and so A is isomorphic to itself.

Symmetric: Suppose that $A \cong B$. This then means there exists a bijection, say $f : A \to B$, between the two sets. Exercise (2.3) proved that the inverse of f, $f^{-1} : B \to A$ is also a bijection, and so this then means that $B \cong A$.

Transitive: Suppose we have $A \cong B$ and $B \cong C$. This then means we have two bijections $f : A \to B$ and $g : B \to C$. Exercise (2.3) also demonstrated that the composition of bijections is a bijection, and so $g \circ f : A \to C$ is a bijection and we have $A \cong C$.

□

Exercise 2.5. Formulate the notion of *epimorphism*, in the style of the notion of *monomorphism* seen in §2.6, and prove a results analogous to Proposition 2.3, for epimorphisms and surjection.

Solution An epimorphism can be defined as follows: A function $f : A \to B$ is an *epimorphism* if the following holds:

for all sets Z and all functions $\alpha', \alpha'' : B \to Z$

$$\alpha' \circ f = \alpha'' \circ f \implies \alpha' = \alpha''$$

The analogous proposition we wish to prove is that *A function is surjective if and only if it is an epimorphism*

($\implies$) Suppose $f : A \to B$ is surjective. Then f has a right inverse. For any functions $\alpha', \alpha'' : B \to Z$ we have

$$\alpha' \circ f = \alpha'' \circ f \implies \alpha' \circ f \circ g = \alpha'' \circ f \circ g \implies \alpha' \circ \mathrm{id}_A = \alpha'' \circ \mathrm{id}_A \implies \alpha' = \alpha''$$

so f is an epimorphism.

($\impliedby$) Converse suppose that f is an epimorphism. Then define $\alpha', \alpha'' : B \to \{0, 1\}$ by the following:

$$\alpha'(b) = \begin{cases} 1 & b \in \mathrm{im} f \\ 0 & b \notin \mathrm{im} f \end{cases}, \alpha''(b) = 1$$

Then clearly we have that $\alpha' \circ f = \alpha'' \circ f$, and by the definition of f being an epimorphism we must have $\alpha' = \alpha''$. This can only be true if $B = \mathrm{im} f$ which means that f must be surjective.

□

Exercise 2.6. With notation as in Example 2.4, explain how any function $f : A \to B$ determine a section of π_A.

Solution Recall that a section is just a right inverse, so we want to find a function $g : A \to A \times B$ such that

$$\pi_A \circ g = \mathrm{id}_A$$

Since $\pi_A((a, b)) = a$, we want to construct g such that $g(a) = (a, b)$ where b is any arbitrary element of B that is not important to us. However, since we want our section to be defined by f we can set $g(a) = (a, f(a))$. This means g is determined by f, and by construction we have that g is a section of π_A.

□

Exercise 2.7. Let $f : A \to B$ be any function. Prove that the graph Γ_f of f is isomorphic to A.

Solution Whenever encountering a question like this, I think its more fruitful to design a function going from the smaller set, A, to the bigger set Γ_f. This is largely due to the fact that when going the other way, you are losing some information, and in cases like this it can be hard to make the function injective. We will construct our function as follows:

$$g : A \to \Gamma_f, a \mapsto (a, f(a))$$

We show that this is a bijection.

Injective: Suppose that $g(a) = g(a')$. This must then mean that $(a, f(a)) = (a', f(a'))$, implying that $a = a'$.

Surjective: for any $(a, f(a)) \in \Gamma_f$ we have that $g(a) = (a, f(a))$ and so g is clearly surjective.

We have constructed a bijection between the two sets, and so we have $A \cong \Gamma_f$.

□